

flight software architecture design.

→ NASA cFs :- Core flight software.

① industry standard for learning modular space architecture.

→ F Prime (F') JPL :-

① component driven software framework.

② Ingenuity mars helicopter.

③ Website contains tutorials, architectural design docs.

→ CCSDS :- Consultative committee for Space Data Systems (CCSDS).

→ Refer EPET 400 : Spacecraft mission design {open source textbook}.

→ Understand the telemetry parameters.

① spacecraft modes:-

① Launch

② deployment

③ Commissioning: Early opⁿ phase; systems are checked

④ Nominal

⑤ safe

⑥ Maintenance

→ ^{**} Telemetry threshold values varies w^ot operational modes.

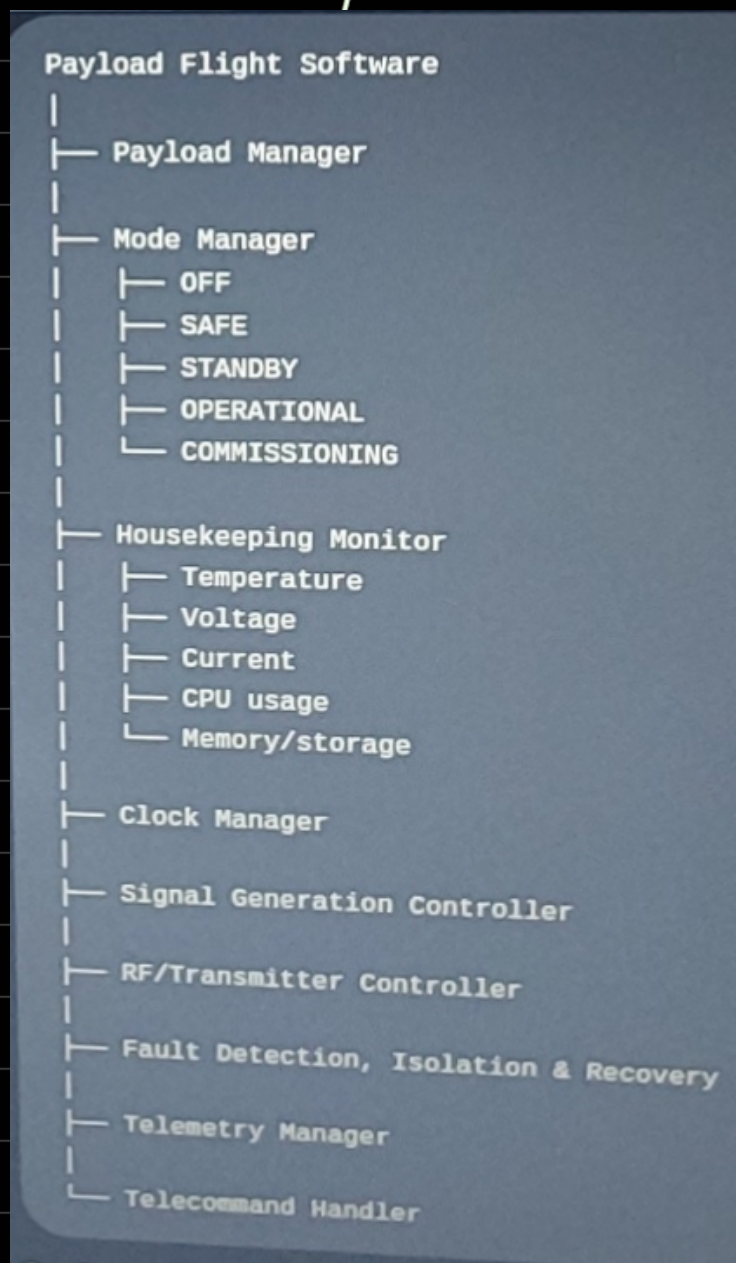
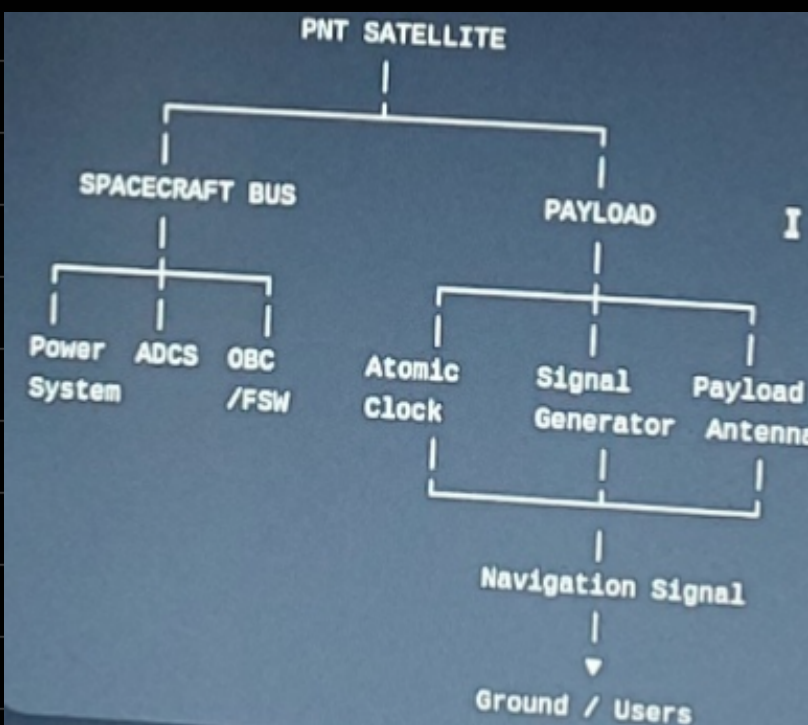
eg:- during commissioning, CPU = 90%.

is expected as engg test hardware.

But, during nominal mode, it may trigger warning.

→ Layman understanding:-

① payload = "mission equipment"
 satellite bus = "platform that supports mission op."



→ Based on my understanding; Almost all telemetry parameters will have both upper & lower threshold limits. It needs to operate within the limits. Any deviations indicates malfunction.

→ Watchdog Counter :- Mechanism that check whether software is still operating correctly.

① The specific number defined to it has no universal meaning. It could represent anything.

→ "solution type: PPP" → Precise point positioning.

→ "clock-bias-ns" → difference b/w receiver's clock and the reference time.

→ Angular rate parameters w/ "gyro" indicate the signs of tumbling and other rotational operations.

→ Transition from nominal to safe mode even for tiny warning is not ideal as it reduces the mission availability. Instead Automatic recovery layer needs to be implemented. However, this introduces risks.

Intuition:- split critical threshold into two zone.

① If parameter > critical-low $\Rightarrow$ Automatic repair mode.

② If parameter > critical-high $\Rightarrow$ SAFE mode and diagnostic.

→ GNSS receivers, OD

→ High rate data path $\Rightarrow$ less code modularity

→ low rate data path $\Rightarrow$ more code modularity

→ FDIR, HK

→ 3.75 - 7.5 cm

→ X band $\rightarrow$ 8-12 GHz $\rightarrow$ High rate mission data
S band $\rightarrow$ 2-4 GHz $\rightarrow$ TT & C
Ku band $\rightarrow$ 12-18 GHz $\rightarrow$ High capacity comms.
Ka band $\rightarrow$ 26-40 GHz $\rightarrow$ V-High " "

→ CCSDS:- protocol to transfer data b/w space system and receives.

